Barath R

Embedded Systems & Firmware Engineer | RISC-V, RTOS & IoT Systems

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PROFESSIONAL SUMMARY

Third-year Electronics & Communication Engineering student and Embedded Firmware Engineer specializing in bare-metal systems, RTOS architecture (FreeRTOS & Zephyr RTOS), and hardware bring-up. Experienced in developing close to the silicon -- from register-level RISC-V trap handlers to peripheral device drivers (I2C, SPI, UART, DMA), sensor fusion, and edge IoT telemetry pipelines. Seeking an Embedded Systems Engineer internship/entry-level role to build high-reliability, real-time connected hardware products.

EDUCATION

B.E. Electronics and Communication Engineering 2024 -- 2028 (Expected)
Rajalakshmi Institute of Technology, Chennai | *Currently in 3rd year* CGPA: 8.65/10

Higher Secondary Certificate (HSC) -- Class XII 2023 -- 2024
Montfort Matriculation Higher Secondary School, Chennai | *Tamil Nadu State Board* Percentage: 95%

Secondary School Leaving Certificate (SSLC) -- Class X 2021 -- 2022
Montfort Matriculation Higher Secondary School, Chennai | *Tamil Nadu State Board* Percentage: 94%

TECHNICAL SKILLS

Embedded & Firmware: Bare-Metal C/C++, FreeRTOS, Zephyr RTOS, ESP-IDF (v6), RISC-V Assembly, Device Drivers, BSP, HAL, ISR, DMA, Timers, PWM, ADC, Low-Power Modes

Hardware & Microcontrollers: ESP32-C6 (RISC-V), ESP32, ESP32-CAM, Raspberry Pi 5, Arduino UNO, Altera DE2 FPGA, Sensor Fusion (6-DOF IMU, Accelerometer, Gyroscope)

PCB Design & EDA Tools: Printed Circuit Board (PCB) Design, KiCad, Proteus, Schematic Capture, 2/4-Layer Layout, Routing, DRC/ERC, Gerber & BOM, Soldering & Board Bring-up

Protocols & Interfacing: I2C, SPI, UART/USART, CAN Bus, GPIO, Memory-Mapped I/O (MMIO), MQTT, LoRaWAN, ESP-NOW, WebSockets, HTTP/REST

Programming Languages: Embedded C, C (C99/C11), C++ (C++17), Python, Java, SQL, RISC-V Assembly, Bash

Cloud & IoT Platforms: HiveMQ, ThingSpeak, Firebase, BlynkIoT, Google Cloud Platform (GCP), AWS IoT Core

Developer Tools & Lab Equipment: Git, GitHub Actions (CI/CD), VS Code, PlatformIO, Linux/Bash, GDB, OpenOCD, JTAG/SWD, Oscilloscope (DSO), Logic Analyzer, Multimeter, CST Studio, Keysight ADS

TECHNICAL PROJECTS

Bare-Metal RISC-V Trap Handler & Context Switcher *ESP32-C6 / Bare-Metal*
Tech Stack: RISC-V Assembly, C, Direct MMIO, Hardware Timers, Interrupt Controller

- Architected a bare-metal trap handler and preemptive context switcher in assembly with 0-byte RTOS overhead, achieving $<$ 2.4 us context-switch latency across 32 registers
- Configured direct Memory-Mapped I/O (MMIO), machine timer interrupts (MTIE), and trap vectors (mtvec) for sub-microsecond hardware interrupt service routine (ISR) execution

ESP-Drone -- Autonomous Flight Controller & Firmware *ESP32-C6 / Quadcopter Firmware*
Tech Stack: C/C++, ESP-IDF, FreeRTOS, IMU Sensor Fusion, PID Control, ESP-NOW / Wi-Fi, MCPWM

- Engineered real-time flight control firmware in C/C++ on ESP32-C6 RISC-V MCU, achieving 500 Hz closed-loop attitude stabilization with $<$ 1% loop jitter via FreeRTOS
- Implemented 6-DOF IMU sensor fusion with complementary filtering, maintaining +-1 deg orientation accuracy, 20 kHz motor PWM, and $<$ 5 ms ESP-NOW wireless telemetry

Dual-Axis Solar Tracker *Embedded Control System*
Tech Stack: Servo Motors, LDR Sensor Bridge, Closed-Loop Control, C/C++

- Constructed a closed-loop dual-axis solar tracking system, increasing photovoltaic energy capture by 28% over static panels through analog differential sensing
- Fabricated motor driver and power regulation circuits; programmed dual-servo closed-loop tracking algorithms across 180 deg azimuth and 90 deg elevation with low-power sleep modes

Digital Twin for Industrial Monitoring *5G-Integrated IoT System*
Tech Stack: ESP32, Sensor Fusion, 5G Telemetry, Python, Real-Time Dashboard

- Architected an industrial edge IoT digital twin connecting multi-sensor ESP32 edge nodes with 5G cellular telemetry, sustaining 99.8% data transmission uptime
- Engineered a sub-100 ms edge-to-cloud sensor ingestion pipeline and real-time dashboard for predictive thermal, vibration, and current anomaly detection

Conformal Helmet-Integrated Antenna Array *RF & Defense Hardware*
Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band, Flexible PCB Substrate, CST Studio, ADS

- Engineered a flexible microstrip patch antenna array conformal to curved helmet geometry, achieving 45% profile reduction and S11 return loss $<$ -10 dB across UHF/L-band
- Simulated 3D radiation patterns and gain in CST Microwave Studio and Keysight ADS, integrating multi-layer copper-clad polyimide with high-isolation RF shielding

TECHNICAL EXPERIENCE & LEADERSHIP

Executive Technical Organizer & Firmware Lead 2024 -- Present
IETE Students' Forum (ISF) Chennai, India

- Spearheaded technical hackathons and hands-on embedded systems workshops, mentoring 500+ student engineers across firmware development, sensor interfacing, and MCU bring-up
- Led end-to-end engineering of 10+ embedded and IoT hardware prototypes, spanning custom PCB bring-up, bare-metal firmware, and real-time cloud dashboards